

Marsh Specialty

Renewable energy: Battery energy storage systems (BESS)





One of the critical challenges of the energy transition is addressing the intermittency of renewable energy generation. As we reduce our reliance on thermal baseload generation – how do we power our homes and businesses when the wind stops blowing and the sun isn't shining?

Renewable energy can be efficiently stored in utility scale battery systems, and power released to the grid when required. This optimization of energy output to the grid means that renewable energy projects can provide power at both peak and non-peak times.

Increased storage capacity and rapidly declining costs of the battery units are driving a global rise in demand. A survey conducted by Bloomberg predicts that by 2030, demand for Lithium-ion (Li-ion) battery capacity will have increased to 9,300 gigawatt hours (GWh) globally. Battery units are most likely to be integrated with solar or wind farms.

Although Li-ion batteries are used in many applications from phones to electric cars, insurers still consider them a prototypical technology when integrated at grid scale. With limited loss data, and costly claims in a number of regions, insurance markets factor the unknown risks of new technology into their capacity, pricing and policy terms. Focus areas for investors and insurers in relation to potential risks include:

Thermal runaway fires

Thermal runaway is a chain reaction where a damaged battery releases energy in the form of heat. Left unchecked the heat can damage adjacent cells, or cause a fire. Large-scale battery fires have occurred in almost every region, notably South Korea, Europe and US. The fires have highlighted that 'cascading' thermal runaway is not unique to a particular manufacturer or design, but that the hazard is inherent in the technology.

Insurance market capacity

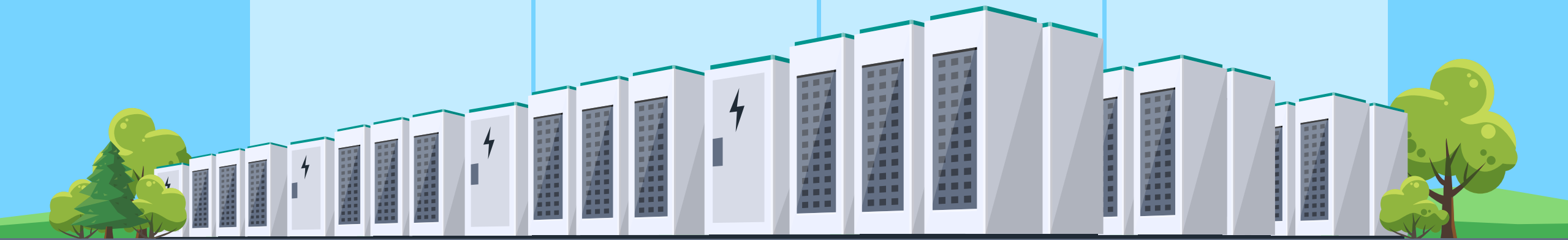
Because energy storage facilities have not yet been adopted at scale across the renewable energy industry, insurance markets generally consider it prototypical technology. Insurers have limited loss data and manufacturing knowledge; and installation and operating standards and best practices vary across regions. With rapidly evolving loss prevention standards, and many unquantified risks, there may be limited insurance market capacity for BESS projects within some regions.

Collateral damage, ensuing loss

The design and configuration of BESS modules is key to containing risk of thermal runaway fires. When assessing the insurability of an installation, insurers will need to be satisfied that there is sufficient separation within the storage modules and between critical infrastructure such as site transformers and substations. Insurers will be focused on ensuring that should an incident occur, the loss and damages can be minimized.

Business interruption

Minimizing single points of failure is key to avoiding contingent business interruption and impact on revenue. Insurers will typically favor projects that route power generated from the renewable energy facility to the grid separately from the BESS, safeguarding the operations and revenue of one component in the event of a loss affecting the other.



A global network of specialists with deep expertise to match your ambitions

Marsh Specialty's global team of renewable energy experts includes insurance advisors, risk engineers, brokers, claims advocates, project finance specialists and former industry risk managers. We use meaningful, relevant data, proprietary modeling tools, and sophisticated analytics to:

- Give you increased control over your short and long-term risks.
- Design cost effective and comprehensive risk transfer programs.
- Engage specialist, global insurance markets.
- Manage complexity at every stage of your project.
- Optimize your total cost of risk.



Risk mapping

From detailed project risk registers and supply chain mapping to on-site engineering and credible loss scenario modeling, we leverage our expertise in BESS technologies to provide our clients with unique insights and tailored risk mitigation strategies.



NatCat analysis

Our approach to mapping natural catastrophe exposures ranges from baseline analysis to precision modeling. Integrated with our engineering expertise, the outputs of our models ensure our clients make informed insurance buying decisions.



Stakeholder management

We support clients in their negotiations with all the major stakeholders, including contractors, lenders and their advisors, to ensure insurance requirements are developed for the benefit of the project.



Access to global insurance markets

Our globally connected team secures capacity across traditional, renewable energy, and alternative specialty insurance markets to deliver the most effective insurance program.



Bespoke policy wording

Our bespoke wording aims to provide best in market coverage, and seamless cover across project phases. It is accepted by major insurers, lenders and their independent advisors.

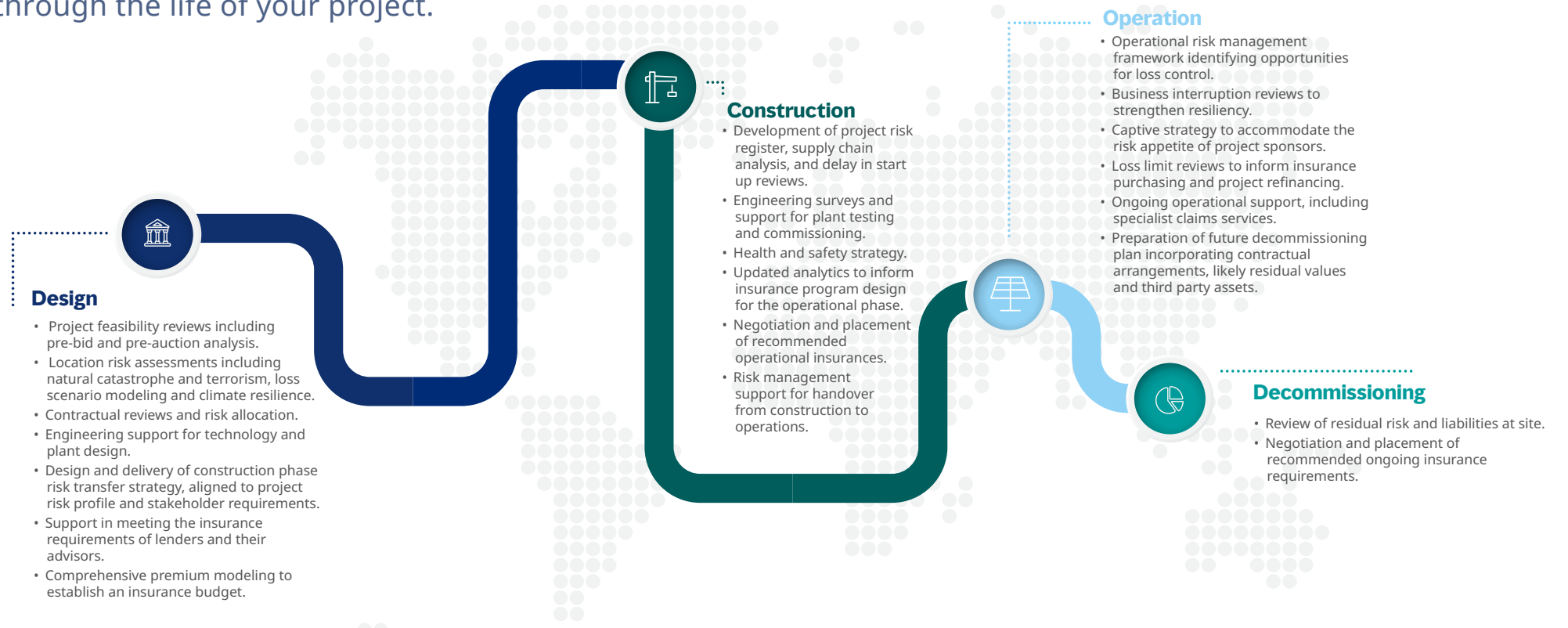


Efficient claims management

Our claims experts implement a transparent, project management approach, engaging effectively with stakeholders to ensure an efficient settlement process to the expected level of quantum.

Providing you with a tailored approach, and comprehensive solutions at every stage

We support you through each project milestone to optimize your risk transfer strategy. We understand that every project is unique, and work with you to design the most effective risk management and insurance program through the life of your project.



A core suite of insurances to protect your assets and your business



Construction

- Construction all risks (CAR)
- Delay in start-up (DSU)
- Third party liability
- Sabotage and terrorism
- Marine cargo



Operation

- Property all risks
- Business interruption
- Third party liability
- Sabotage and terrorism physical damage and business interruption



Additional

- Cyber
- Environmental impairment liability
- Professional indemnity
- Political risk
- Directors' and officers' liability
- Transactional and credit risk
- Tax insurance solutions
- Parametric solutions
- Performance guarantee
- Commissioning risk
- Compulsory local insurances

We are where you are

We regularly bring global resource to local clients. Around the world, our team has designed risk management programs and secured insurance for over 2,000 renewable energy projects.

GLOBAL

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OVER
400

globally coordinated team of renewable energy experts located across eight geographical hubs.

OVER
650 GIGAWATTS

of renewable energy in our portfolio.

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